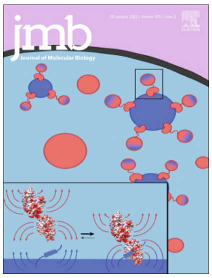
Methods Note



BASIN: A Semi-automatic Workflow, with Machine Learning Segmentation, for Objective Statistical Analysis of Biomedical and Biofilm Image Datasets

Timothy W. Hartman [1#], Evgeni Radichev [1#], Hafiz Munsub Ali [1†], Mathew Olakunle Alaba [1], Mariah Hoffman [1], Gideon Kassa [1‡], Rajesh Sani [2], Venkata Gadhamshetty [3], Shankarachary Ragi [4], Shanta M. Messerli [1,5,6], Pilar de la Puente [5], Eric S. Sandhurst [1], Tuyen Do [1], Carol Lushbough [1] and Etienne Z. Gnimpieba [1*]

*1 - **Biomedical Engineering Department,** University of South Dakota Sioux Falls, 4800 N Career Avenue, Sioux Falls, SD 57107, United States*

*2 - **Chemical and Biological Engineering Department,** South Dakota School of Mines and Technology, 501 E St. Joseph Street, Rapid City, SD 57701, United States*

*3 - **Civil and Environmental Engineering Department,** South Dakota School of Mines and Technology, 501 E St. Joseph Street, Rapid City, SD 57701, United States*

*4 - **Electrical Engineering Department,** South Dakota School of Mines and Technology, 501 E St. Joseph Street, Rapid City, SD 57701, United States*

*5 - **Cancer Biology and Immunotherapies Group,** Sanford Research, 2301 E 60th Street North, Sioux Falls, SD 57104, United States*

*6 - **Department of Biology and Microbiology,** South Dakota State University, Brookings, SD 57006, United States*

Correspondence to Etienne Z. Gnimpieba:*Biomedical Engineering Department, University of South Dakota Sioux Falls, 4800 N Career Avenue, Sioux Falls, SD 57107, United States. *Etienne.Gnimpieba@usd.edu (E.Z. Gnimpieba)*
https://doi.org/10.1016/j.jmb.2022.167895
Edited by Rita Casadio

Abstract

Micrograph comparison remains useful in bioscience. This technology provides researchers with a quick snapshot of experimental conditions. But sometimes a two- condition comparison relies on researchers' eyes to draw conclusions. Our Bioimage Analysis, Statistic, and Comparison (BASIN) software provides an objective and reproducible comparison leveraging inferential statistics to bridge image data with other modalities. Users have access to machine learning-based object segmentation. BASIN provides several data points such as images' object counts, intensities, and areas. Hypothesis testing may also be performed. To improve BASIN's accessibility, we implemented it using R Shiny and provided both an online and offline version. We used BASIN to process 498 image pairs involving five bioscience topics. Our framework supported either direct claims or extrapolations 57% of the time. Analysis results were manually curated to determine BASIN's accuracy which was shown to be 78%. Additionally, each BASIN version's initial release shows an average 82% FAIR compliance score.

Introduction

Bioimage analysis remains a core life science research tool. Unlike other analysis tools, bioimaging is useful to track changes in objects' positions, and morphologies over time and between various experimental conditions.[1] Additionally, modern microscopes allow for the semi-automatic capture of multiple two- and three-dimensional images within one sample. The growth of open-source code has helped make image analysis tools more popular,[2–3] while the development of deep learning algorithms has led to dramatic improvements in the efficacy of bioimage restoration, registration, and classification, as well as object detection, segmentation, and tracking.[4,5] These modern advancements lay the groundwork for the automation of bioimage analysis, which is crucial for the future of biological research. Scientists will be able to leverage next generation tools to expediate the analysis process as opposed to spending time identifying and counting objects of interest and performing image comparisons individually by hand.

The development of state-of-the-art image analysis tools follows the need of a growing diversity of image acquisition techniques and digital image processing (DIP) tasks.[6] These tasks include segmentation, object detection, and feature extraction that are best executed with high resolution, low noise images.[6] The most popular open-source tools to address DIP tasks include the ImageJ family,[6–8] CellProfiler,[9] and Icy.[10] These tools perform a wide variety of DIP tasks 'on-the-fly' or by reusable, shareable workflows. Capturing multiple images per sample can improve statistical power but analyzing them without the use of semi-automatic functions or macros is time-consuming. Additionally, basic hypothesis testing is not directly implemented in some tools. In the case of CellProfiler, interoperability with R is required. Therefore, many bioimage analysis tools remain complex and difficult for bioscience, bioengineering, and environmental biotechnology researchers to deploy effectively.

Here we explore the Bioimage Analysis, Statistic, and Comparison (BASIN) workflow for comparing bioimage datasets using descriptive and inferential statistics. BASIN allows for bioimage comparison at various dataset sizes: from two images with a simple web interface (tryBASIN), to a mid-sized dataset comparison using either simple segmentation (BASINlite) or machine learning (BASIN-ML), to analyzing a larger dataset using simple segmentation with the BASIN Notebook in Kaggle. BASIN uses the R package 'EBImage' for image processing and feature extraction.[11] Upon analysis completion, all BASIN versions allow the user to generate a provenance report and an output folder containing all data extraction and analysis results. The provenance report keeps track of all steps in the analytic workflow process to improve reproducibility and quality control. The output is ready to share with the community using common data sharing platforms in accordance with the Findability, Accessibility, Interoperability, and Reusability (FAIR) guidelines.[12]

Results and Discussion

Bioimaging technology allows researchers to provide the community with a quick visual snapshot of experimental conditions for comparison. BASIN provides an easy, objective, reproducible bioimage comparison which leverages inferential statistics. Several key data points are captured including images' object counts, net and mean pixel intensities, net and mean object surface areas, plus a variety of other potentially useful data. Hypothesis testing is performed on mean object intensities and surface areas using the statistical power of the R programming language. Researchers gain knowledge and insight into a wide range of topics, in diverse areas such as drug protein marker response, the significance of cell population change, and changes in cell morphology. The BASIN workflow consists of five core modules including image upload, feature extraction, statistical analysis, visualization, and report generation. Here, we used our software to process 498 image pairs involving five different bioscience topics. These topics include microbial biofilms, biomaterials, cancer, diabetes, and SARS-CoV-2, and the images include confocal and scanning electron micrographs of varying qualities: from internet-sourced images to high-resolution micrographs. The diverse datasets are summarized in Figure 1. Our framework supported either direct claims or extrapolations 57% of the time. Analysis results were manually curated to determine the accuracy of BASIN. Based on our curation, our framework showed an analysis accuracy of 78%. In addition, each BASIN object's initial version within our toolkit was assessed and shows an average Findability, Accessibility, Interoperability, and Reusability (FAIR) compliance score of 82%.

The purpose of our evaluation was to assess our tools' ability to generate meaningful results. Specifically, we evaluated BASINlite and BASIN Notebook on all datasets, which are the two versions capable of processing hundreds of images in one session. Our remaining applications, tryBASIN and BASIN-ML, were tested on a subset of the full dataset and were not used for performance evaluation.

Performance evaluation

To validate the reproducibility of the BASIN framework, we tested our workflow on the

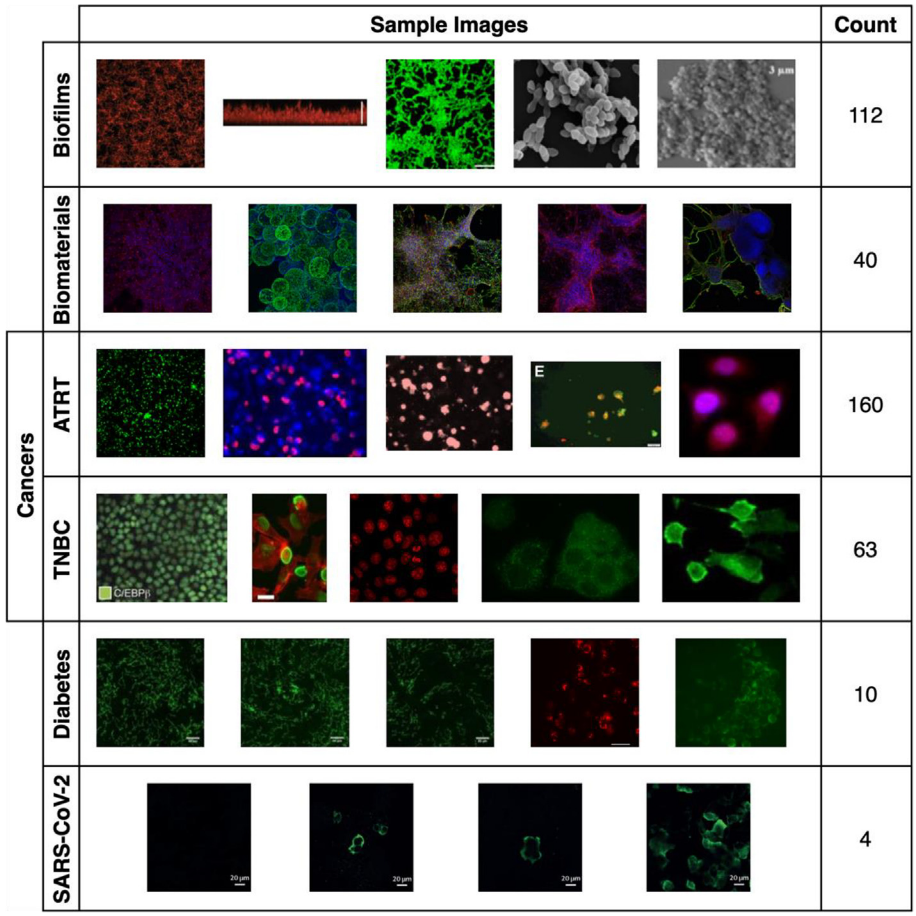


Figure 1. Image dataset diversity overview. A diverse image collection was gathered from various sources for five bioscience topics. The images shown here: biofilms,[26–29] biomaterials courtesy of Eric Sandhurst, ATRT,[30–34] TNBC,[35–39] diabetes,[40–41] and SARS-CoV-2.[42] Most of the images were manually cropped from published article figures. Two collections contain original, high-resolution images courtesy of Hoffman et al[30] and Eric Sandhurst. While the image collection is primarily confocal micrographs, the biofilms set has a mix of confocal and SEM images. Note that some images from published article figures are annotated with scale bars and other labels. These annotations would be recognized by BASIN as objects.

complete evaluation dataset using both BASINlite and a custom Kaggle notebook. In Kaggle, the entire dataset was uploaded and run in one session. For BASINlite, multiple sessions were required due to the limitations outlined in Materials and Methods. The evaluation tested BASIN's ability to read images, perform basic segmentation using the Otsu method,[13] use the analysis table CSVs to divide images accordingly, perform t-tests on image pairs if specified in the analysis tables, and generate a spreadsheet of analysis results for each image pair. After the Kaggle session, the output folders were checked for completion. These results were cross-referenced with the reports generated by BASINlite to ensure consistency between the two evaluation frameworks. Additionally, BASINlite was used to assess the quality of BASIN's visualizations and generate a summary report.

We found that BASINlite could analyze and provide output on every image set except a few whose images could not be processed by EBImage's Compute Features functions. Additionally, BASINlite could process between 25–30 MB of images at once on a 64-bit machine with 8 GB of RAM running R version 4.0.3. BASIN Notebook could process the entire dataset in one session provided the images could be read by EBImage.

The average time required to run the entire analysis workflow on BASINlite for a single pair of ATRT cancer images with pixel dimensions 512×512 and average size of 150 KB was under 90 seconds, assuming a pre-made analysis table. The breakdown of CPU wait times for the most computationally intensive tasks is as follows: thresholding, color labeling, and object feature extraction took 10 seconds; statistical analysis took less than 5 seconds; generating the BASIN report took 15 seconds. All remaining time is considered "active user time" because the user is either interacting with the application or viewing the intermediate results. For a set of 20 images from the same dataset with the same properties, the average analysis time was under 200 seconds. Thresholding, color labeling, and object feature extraction took 35 seconds, statistical analysis took 8 seconds, and generating the report took 85 seconds.

Our evaluation of BASIN reveals the potential for mass image processing and analysis with proper memory management, which the current BASIN versions lack. Compiling the image dataset revealed the challenge of acquiring high-quality images from published research which normally requires correspondence with those papers' authors. For this purpose, we are driven to develop BASIN in compliance with the FAIR guidelines.[12]

Accuracy evaluation

To assess the accuracy of BASIN, the workflow's output was validated against multiple manual curations and papers' claims to check if the image supports the claim and if the features extracted are reasonable within the scope of the research (Supplementary Table 1).

In the first assessment, BASIN results were compared to biological knowledge of the underlying processes being imaged. Each image pair had a claim associated with it that used objects' pixel intensity, count, or surface area to match most closely what was either claimed by or could be extrapolated from the published article text. "Direct" claims are ones made by the authors about that pair of images. "Supported extrapolations" are claims where the authors have other published data to back that claim, and which may or may not correspond to images supporting that claim. For example, the authors may have used flow cytometry to count cells, and images were added to help visualize the difference between two conditions. "Unsupported extrapolations" are statements that may be true about an image pair, but there is no published data to support it. For instance, one might expect a higher cell count on day three compared to day zero, but the authors made no explicit claim.

In addition to claims based on biological knowledge, BASIN results were validated against expert visual inspection of two-image pairs. For this evaluation, 180 image pairs were selected and differences assessed between the manually curated results and BASIN results (Figure 2). BASIN's accuracy for all comparisons was evaluated using precision, recall, and F-measure metrics.[14] Given a true positive $TP = 128$, false positive $FP = 1$, and false negative $FN = 38$, the BASIN framework had an overall precision of 0.992, recall of 0.771, and F-measure of 0.868. BASIN's high precision and moderately high recall indicate that the implemented hypothesis testing approach may tend slightly conservative, which is a useful starting place for many bioscience applications.

Two neural network architectures were used to improve segmentation in complex image datasets. Cellpose[15] and U-Net[16] have proven to be the gold standards in cell segmentation, which allow the user to add more knowledge on pre-trained models for specialized datasets than a generalized model (e.g., add new dataset collection to U-Net) or integrate more complex segmentation tasks using the multi-task mechanism (e.g., add new task from complex dataset in Cellpose). Previous research has assessed the segmentation performance of both Cellpose and U-Net, where Cellpose outperforms U-Net architectures "on complex images in terms of F1 scores, while the U-Net architectures achieve overall higher mean Dice scores".[17] The Convolutional Neural Network (CNN) implemented in a pre-trained U-Net has been shown to reach a precision of up to 84.1% and recall of 79.3%.[17] The multi-task CNN implemented in a pre-trained Cellpose network has been shown to reach a precision of up to 90.8% and recall of 82.9%.[17] Qualitative inspection demonstrates the advantage of these deep neural networks over Otsu thresholding in complex tasks such as biofilm dataset segmentation, as shown in Figure 3. However, comparing the segmentation performance of each using quantitative measures is beyond the scope of this work. These results highlight that machine learning plays a key role in the most accurate segmentation of cells and other microorganisms.[4]

Fairness assessments

The BASIN workflow was evaluated to check the FAIR compliance at the software level as well as at the dataset and analysis result levels.[18] We evaluated the FAIR compliance of the BASIN project including its tools and the dataset used to evaluate BASIN version 1.0. The latest BASIN project assessment in FAIRshake.cloud indicated an average 82% FAIRness score across all relevant digital objects.[18] A breakdown of FAIRness scores by digital object is shown in Table 1. FAIRshake insignias are shown in Table 1 and are described in Supplementary Figure 3.

Materials and Methods

Input, upload, and segmentation

Users can upload images to BASIN from a local computer/disk or from online databases. In all BASIN applications users can upload images using the graphical user interface. Users must also upload a CSV file detailing the experimental groups and analysis parameters (see Supplementary Figure 2). A template for this file can be downloaded directly from BASIN and modified to fit the user's analysis objective. The initial BASIN version focuses on two-dimensional

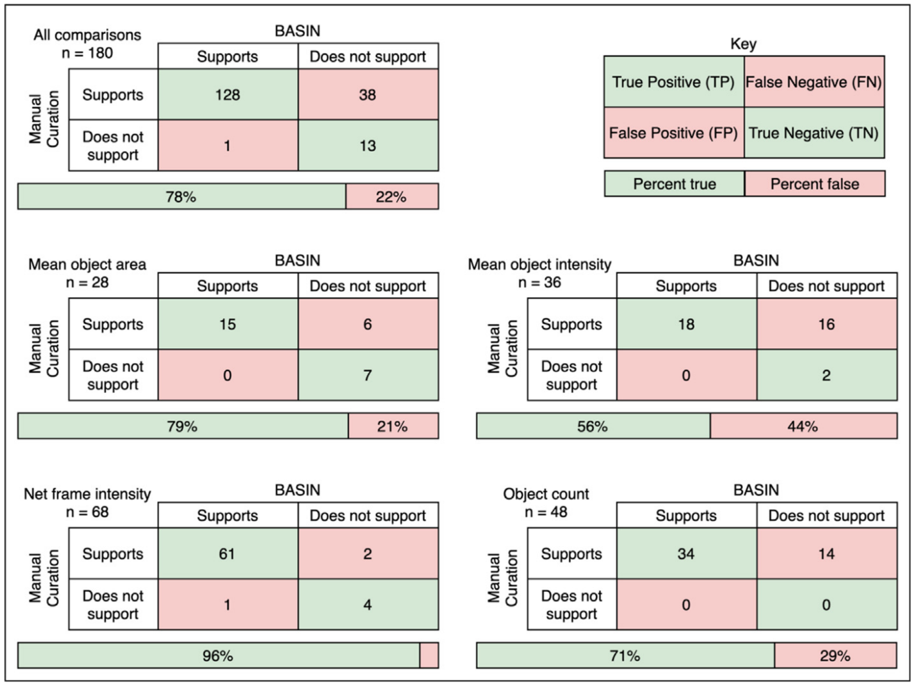


Figure 2. Confusion matrices for various feature comparisons. There was a total of 180 two-image comparisons where BASIN was able to analyze the image pair and the manual curator could support or reject the claim or extrapolation using their eyes. Of these 180 comparisons, 28 were comparing mean object areas, 36 were comparing mean object intensities, 68 were comparing net frame (channel) intensities, and 48 were comparing object counts.

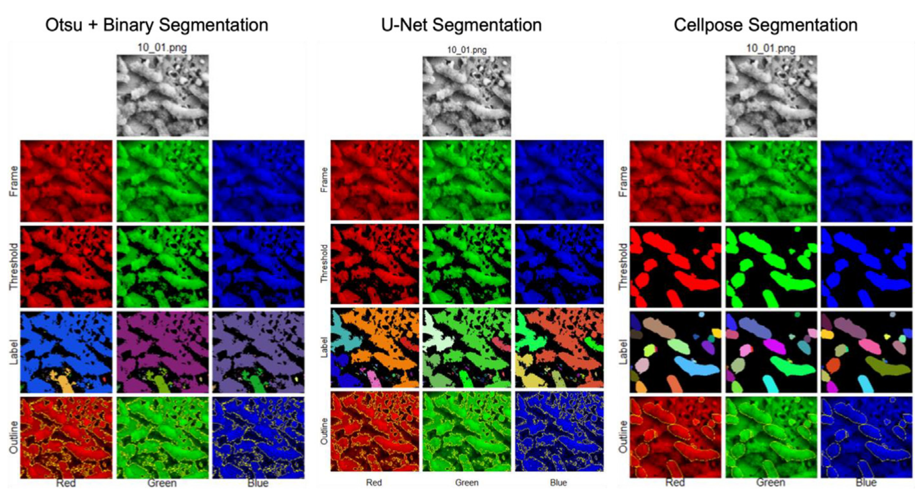
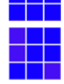
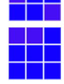
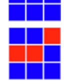
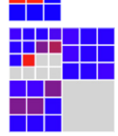


Figure 3. Performance comparison of BASIN segmentation methods. An SEM image of biofilm was used to qualitatively test performance.[43] For all three method types, the original image is split into red, green, and blue frames. The frames are colorized here for ease of understanding. Objects in each frame are segmented either using one of three methods: thresholding (e.g., Otsu), a machine learning method such as CNN leveraging pre-trained U-Net, or using multi-task CNN with a pre-trained Cellpose model. Each object in a frame is given a unique label, then the 'EBImage' R package extracts morphological data from each object. BASIN uses this data to perform inferential statistics on the objects of each frame.

(2D) images in a limited format capacity, so a 3D image must be sliced to comply to this limitation. Once uploaded, the image dataset is split into three groups, one for each RGB color frame, and converted into a numerical matrix representation for feature extraction.

Table 1 FAIRshake assessments and scores.

Digital Object (verbatim)	FAIRness Score	Insignia
tryBASIN FAIR	58%	
BASIN Lite v1.0	81%	
BASIN Notebook v1.0	83%	
BASIN Dataset	100%	
BASIN Analysis Report v1.0	92%	
BASIN Project Website	94%	
BASINlite v1.0 Output Folder	67%	
BASIN Project (average score)	82%	

All BASIN packages implement automatic thresholding using the 'autothresholdr' R package,[19] which functions as an R port of ImageJ's "AutoThreshold" plugin. A description of automatic thresholding methods can be found at https://imagej.net/plugins/auto-threshold. In addition to auto-thresholding, BASIN-ML allows machine learning-based segmentation using Cellpose,[15] U-Net,[16] or a user-uploaded HDF5 model. Researchers can fine-tune or train custom models using their data and maintain their usability in BASIN for analyses. Since all segmentation models are Python-based, BASIN-ML utilizes the 'reticulate' R package[20] to allow interoperability between two languages. Supplementary Table 2 contains a brief description of the segmentation models provided in BASIN-ML.

Bioimage feature extraction

After thresholding and segmentation, various image-level descriptive statistics are computed and stored in one file including net image intensity, object count, sum object intensity, and others. At the object level, standard features are extracted using EBImage's 'computeFeatures' functions and stored along with their descriptive statistics in unique CSV output files for each RGB color frame. These features include pixel intensities, object shape, object image moments, and pixel texture. BASIN results include pre-threshold (as raw) and post-threshold (as cumulative) intensities.[21] The data in these files are further used to present, plot, and perform statistical analysis. Once all detected object features have been computed, users have the option to further threshold their data using the minimum and maximum object sizes, which are specified by total pixel counts. This allows users to manually remove outliers and other undesired objects from the dataset for improved results.

Statistical hypothesis testing

BASIN application adopts hypothesis testing to verify the statistical significance of the supplied image comparisons. Mean values may not represent the true performance of the BASIN application and can be misleading due to random fluctuations in the data. A cross-sample comparison is necessary to determine if the difference between two sample means is significant or simply due to random chance. Therefore, a t-test is used to evaluate the degree of reliability and performance significance in the mean values that are generated by the BASIN application. Performance significance is measured between two mean samples of data. In BASIN, these samples are represented by the "control" and "test" conditions for each experimental group. Since these two samples may arise from entirely different population distributions, they are treated as unpaired samples with unequal variances. Therefore, we conduct a Welch's unequal variances t-test between the "control" and "test" groups for each user-specified experimental group.[22] P-values are computed with the generally acceptable significance level of $\alpha = 0.05$.

In our analysis, the comparison of means is always assumed to be $\mu1$ (control) against $\mu2$ (test). This implies that when the p-value $\leq \alpha$, then there is a statistically significant difference (with a default 95% confidence level) between the sample means of the control group and the test group.[22] Otherwise, the observed difference is not statistically significant. Should a one-sided test be desired, then the sign of the alternative hypothesis will match the comparison $\mu1$ against $\mu2$. For example, a less-than alternative hypothesis will result in the test $\mu1 < \mu2$, where a p-value < 0.05 is statistically significant evidence that the true mean of the control group is less than the true mean of the test group. Users can specify the type of comparison in the experimental design CSV file. Welch's t-test generates tabulated results for two or more data samples, which BASIN converts into a results table with the columns representing alternate hypothesis (Ha), t-statistic, degree of freedom (DF), p-value, lower confidence interval, upper confidence interval, confidence level, etc.

Data visualization and reporting

Data visualization is provided by the 'ggplot2'[23] and 'DT'[24] R packages. These packages provide high-quality, interactive graphs and tables that allow users to observe the distribution of the data, determine the existence of outliers, and compare their sample groups within each experiment. BASIN's Viewer module (Supplementary Figure 2) prompts the user to review each step of the workflow, reanalyze or change analysis parameters if desired, and visualize all steps in their analysis in a single diagram. The report generated by BASIN includes every step in the analysis workflow as presented in the Viewer module in various formats along with a step-by-step analysis folder (Supplementary File 1). This provenance report can be published in any data repository (e.g., Mendeley Data, FigShare) to improve user analysis and add as a reference.

FAIR method

FAIR guiding principles help users and their machines maximize image data "transparency, reproducibility, and reusability".[12] This was considered a data science grand challenge in 2016. In general, these principles indicate a desire by the tool's developers to guide their software in the direction of better data management. To this end, BASIN attempts to follow FAIR principles (Supplementary Figure 3) by providing a consistent workflow for reproducibility, open-source code for transparency, and a software package for reusability.

Availability

BASIN's website is found at https://basin-v10.readthedocs.io/en/latest. The BASIN repository is available in GitHub at https://github.com/bicbioeng/BASIN. The tryBASIN Shiny app is available at https://bicbioeng.shinyapps.io/tryBASIN and on our own server at https://basin.bicbioeng.org/ (recommended). The BASIN project's FAIR assessments are located at https://fairshake.cloud/project/130. The analysis of this paper is available in Mendeley data at https://data.mendeley.com/datasets/t3f3w77fjk/1. Upon publication, the tools will be shared with bioinformatics repositories such as Bio-TDS.org.[25] Dataset and supplementary files, figures, and tables are available at doi: 10.17632/t3f3w77fjk.3. Due to copyright restrictions and intellectual property of certain images, other parts of our evaluation dataset may be obtained by request.

CRediT authorship contribution statement

Timothy W. Hartman: . **Evgeni Radichev:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – review & editing, Visualization, Supervision, Project administration. **Hafiz Munsub Ali:** Validation, Formal analysis, Resources, Data curation, Writing – review & editing, Supervision. **Mathew Olakunle Alaba:** Resources, Data curation. **Mariah Hoffman:** Conceptualization, Methodology, Validation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Gideon Kassa:** Conceptualization, Data curation, Writing – review & editing, Visualization. **Rajesh Sani:** Resources, Data curation, Writing – review & editing. **Venkata Gadhamshetty:** Resources, Data curation, Writing – original draft, Writing – review & editing. **Shankarachary Ragi:** Writing – original draft, Writing – review & editing. **Shanta M. Messerli:** Resources, Data curation, Writing – original draft, Writing – review & editing. **Pilar de la Puente:** Resources, Data curation, Writing – original draft, Writing – review & editing. **Eric S. Sandhurst:** Resources, Data curation, Writing – original draft, Writing – review & editing. **Tuyen Do:** Software, Validation, Formal analysis, Resources, Data curation, Writing – review & editing. **Carol Lushbough:** Validation, Writing – original draft, Writing – review & editing. **Etienne Z. Gnimpieba:** Conceptualization, Methodology, Software, Validation, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Office of Experimental Program to Stimulate Competitive Research of the National Science Foundation under award numbers 1849206 and 1920954. It was also supported by the Foundation of the National Institutes of Health under award number 5P20GM103443-20. Mariah Hoffman is supported by a grant from the National Science Foundation Research Traineeship program, USD-N3 (DGE-1633213). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Science Foundation or the National Institutes of Health.

Appendix A. Supplementary Data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jmb.2022.167895.

Received 12 July 2022;
Accepted 16 November 2022;
Available online 1 December 2022

Keywords:
image comparison;
microscope imaging;

machine learning;
segmentation;
statistical analysis;
FAIR;
Microbe;
biofilm;
cancer

† Present address: Computer Science Department, Binghamton University – SUNY, 4400 Vestal Parkway East, Binghamton, NY 13902, United States.
‡Present address: Materials Engineering Department, Dartmouth College, 15 Thayer Drive, Hanover, NH 03755, United States.
#Timothy W. Hartman, and Evgeni Radichev the first two authors are equal contributors.

References

1. Boquet-Pujadas, A., Olivo-Marin, J.-C., Guillén, N., (2021). Bioimage analysis and cell motility. *Patterns* **2** (1), 100170. https://doi.org/10.1016/j.patter.2020.100170.
2. Web of Science, "'ImageJ' Citation Report." https://www.webofscience.com/wos/woscc/citation-report/47747b24-1300-4e01-af06-a7cc1f32b11a-006148d1?page=1 (accessed Jul. 08, 2021).
3. Eliceiri, K.W. et al, (2012). Biological imaging software tools. *Nat. Methods* **9** (7), 697–710. https://doi.org/10.1038/nmeth.2084.
4. Meijering, E., (2020). A bird's-eye view of deep learning in bioimage analysis. *Comput. Struct. Biotechnol. J.* **18**, 2312–2325. https://doi.org/10.1016/j.csbj.2020.08.003.
5. Hallou, A., Yevick, H.G., Dumitrascu, B., Uhlmann, V., (2021). Deep learning for bioimage analysis in developmental biology. *Development* **148** (18), dev199616. https://doi.org/10.1242/dev.199616.
6. Schindelin, J. et al, (2012). Fiji: an open-source platform for biological-image analysis. *Nat. Methods* **9** (7), 676–682. https://doi.org/10.1038/nmeth.2019.
7. Schneider, C.A., Rasband, W.S., Eliceiri, K.W., (2012). NIH Image to ImageJ: 25 years of image analysis. *Nat. Methods* **9** (7), 671–675. https://doi.org/10.1038/nmeth.2089.
8. Rueden, C.T. et al, (2017). ImageJ2: ImageJ for the next generation of scientific image data. *BMC Bioinf.* **18** (1), 529. https://doi.org/10.1186/s12859-017-1934-z.
9. Carpenter, A.E. et al, (2006). CellProfiler: image analysis software for identifying and quantifying cell phenotypes. *Genome Biol.* **7** (10), R100. https://doi.org/10.1186/gb-2006-7-10-r100.
10. de Chaumont, F. et al, (2012). Icy: an open bioimage informatics platform for extended reproducible research. *Nat. Methods* **9** (7), 690–696. https://doi.org/10.1038/nmeth.2075.
11. Pau, G., Fuchs, F., Sklyar, O., Boutros, M., Huber, W., (2010). EBImage–an R package for image processing with applications to cellular phenotypes. *Bioinformatics* **26** (7), 979–981. https://doi.org/10.1093/bioinformatics/btq046.
12. Wilkinson, M.D. et al, (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Sci. Data* **3** (1), 160018. https://doi.org/10.1038/sdata.2016.18.
13. Otsu, N., (1979). A Threshold Selection Method from Gray-Level Histograms. *IEEE Trans. Syst. Man Cybern.* **9** (1), 62–66. https://doi.org/10.1109/TSMC.1979.4310076.
14. Fawcett, T., (2006). An introduction to ROC analysis. *Pattern Recogn. Lett.* **27** (8), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010.
15. Stringer, C., Wang, T., Michaelos, M., Pachitariu, M., (2021). Cellpose: a generalist algorithm for cellular segmentation. *Nat. Methods* **18** (1), 100–106. https://doi.org/10.1038/s41592-020-01018-x.
16. Ronneberger, O., Fischer, P. & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation BT - Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. pp. 234–241.
17. Kromp, F. et al, (2021). Evaluation of deep learning architectures for complex immunofluorescence nuclear image segmentation. *IEEE Trans. Med. Imaging* **40** (7), 1934–1949. https://doi.org/10.1109/TMI.2021.3069558.
18. E. Gnimpieba, "BASIN Project FAIRshake." https://fairshake.cloud/project/130/ (accessed May 14, 2021).
19. Landini, G., Randell, D.A., Fouad, S., Galton, A., (2017). Automatic thresholding from the gradients of region boundaries. *J. Microsc.* **265** (2), 185–195. https://doi.org/10.1111/jmi.12474.
20. Ushey, K., Allaire, J. & Tang, Y. (2021). reticulate: Interface to 'Python. [Online]. Available: https://cran.r-project.org/package=reticulate.
21. Landini, G. (2021). Auto Threshold, *ImageJ*. https://imagej.net/plugins/auto-threshold (accessed Jul. 08, 2021).
22. Welch, B.L., (1947). The generalization of 'Student's' problem when several different population variances are involved. *Biometrika* **34** (1–2), 28–35. https://doi.org/10.1093/biomet/34.1-2.28.
23. Wickham, H. (2009). *ggplot2*, 1st ed. New York, NY: Springer New York, 2009. https://doi.org/10.1007/978-0-387-98141-3.
24. Xie, Y., Cheng, J., Tan, X., (2021). DT: A wrapper of the javascript library 'DataTables'. *CRAN*.
25. Gnimpieba, E.Z., VanDiermen, M.S., Gustafson, S.M., Conn, B., Lushbough, C.M., (2017). Bio-TDS: bioscience query tool discovery system. *Nucleic Acids Res.* **45** (D1), D1117–D1122. https://doi.org/10.1093/nar/gkw940.
26. Mancera, E. et al, (2020). Evolution of the complex transcription network controlling biofilm formation in Candida species. *bioRxiv*, 1–24. https://doi.org/10.1101/2020.11.08.373514.
27. Sherry, L. et al, (2013). Investigating the biological properties of carbohydrate derived fulvic acid (CHD-FA) as a potential novel therapy for the management of oral biofilm infections. *BMC Oral Health* **13** (1), 47. https://doi.org/10.1186/1472-6831-13-47.
28. Xie, F., Zhang, Y., Li, G., Zhou, L., Liu, S., Wang, C., (2013). The ClpP Protease Is Required for the Stress Tolerance and Biofilm Formation in Actinobacillus pleuropneumoniae. *PLoS One* **8** (1), e53600.
29. Nagraj, A.K., Gokhale, D., (2018). Bacterial Biofilm Degradation Using Extracellular Enzymes Produced by Penicillium janthinellum EU2D-21 under Submerged Fermentation. *Adv. Microbiol.* **08** (09), 687–698. https://doi.org/10.4236/aim.2018.89046.
30. Hoffman, M.M. et al, (2020). Analysis of Dual Class I Histone Deacetylase and Lysine Demethylase Inhibitor Domatinostat (4SC-202) on Growth and Cellular and

Genomic Landscape of Atypical Teratoid/Rhabdoid. *Cancers (Basel)* **12** (3), 756. https://doi.org/10.3390/cancers12030756.

31. Weingart, M.F. et al, (2015). Disrupting LIN28 in atypical teratoid rhabdoid tumors reveals the importance of the mitogen activated protein kinase pathway as a therapeutic target. *Oncotarget* **6** (5), 3165–3177. https://doi.org/10.18632/oncotarget.3078.
32. Kaur, H. et al, (2015). The chromatin-modifying protein HMGA2 promotes atypical teratoid/rhabdoid cell tumorigenicity. *J. Neuropathol. Exp. Neurol.* **74** (2), 177–185. https://doi.org/10.1097/NEN.0000000000000161.
33. Messerli, S.M., Hoffman, M.M., Gnimpieba, E.Z., Bhardwaj, R.D., (2017). Therapeutic targeting of PTK7 is Cytotoxic in atypical teratoid rhabdoid tumors. *Mol. Cancer Res.* **15** (8), 973–983. https://doi.org/10.1158/1541-7786.MCR-16-0432.
34. Jayanthan, A., Bernoux, D., Bose, P., Riabowol, K., Narendran, A., (2011). Multi-tyrosine kinase inhibitors in preclinical studies for pediatric CNS AT/RT: Evidence for synergy with Topoisomerase-I inhibition. *Cancer Cell Int.* **11** (1), 44. https://doi.org/10.1186/1475-2867-11-44.
35. Johansson, J. et al, (2013). MiR-155-mediated loss of C/EBPβ shifts the TGF-β response from growth inhibition to epithelial-mesenchymal transition, invasion and metastasis in breast cancer. *Oncogene* **32** (50), 5614–5624. https://doi.org/10.1038/onc.2013.322.
36. Tate, C.R. et al, (2012). Targeting triple-negative breast cancer cells with the histone deacetylase inhibitor panobinostat. *Breast Cancer Res.* **14** (3), R79. https://doi.org/10.1186/bcr3192.
37. Machowska, M., Wachowicz, K., Sopel, M., Rzepecki, R., (2014). Nuclear location of tumor suppressor protein maspin inhibits proliferation of breast cancer cells without affecting proliferation of normal epithelial cells. *BMC Cancer* **14** (1), 142. https://doi.org/10.1186/1471-2407-14-142.
38. Samartzis, E.P., Noske, A., Meisel, A., Varga, Z., Fink, D., Imesch, P., (2014). The G Protein-Coupled Estrogen Receptor (GPER) is expressed in two different subcellular localizations reflecting distinct tumor properties in breast cancer. *PLoS One* **9** (1), e83296.
39. Bilir, B., Kucuk, O., Moreno, C.S., (2013). Wnt signaling blockage inhibits cell proliferation and migration, and induces apoptosis in triple-negative breast cancer cells. *J. Transl. Med.* **11** (1), 280. https://doi.org/10.1186/1479-5876-11-280.
40. Wang, Q. et al, (Dec. 2010). Mitochondrial Dysfunction and Apoptosis in Cumulus Cells of Type I Diabetic Mice. *PLoS One* **5** (12), e15901.
41. Yorek, M.S., Coppey, L.J., Shevalye, H., Obrosov, A., Kardon, R.H., Yorek, M.A., (2016). Effect of treatment with salsalate, menhaden oil, combination of salsalate and menhaden oil, or resolvin D1 of C57Bl/6J type 1 diabetic mouse on neuropathic endpoints. *J. Nutr. Metab.* **2016**, 1–11. https://doi.org/10.1155/2016/5905891.
42. Chu, H. et al, (2020). Comparative tropism, replication kinetics, and cell damage profiling of SARS-CoV-2 and SARS-CoV with implications for clinical manifestations, transmissibility, and laboratory studies of COVID-19: an observational study. *Lancet Microbe* **1** (1), e14–e23. https://doi.org/10.1016/S2666-5247(20)30004-5.
43. Rahman, M. H. Duckworth, J., Ragi, S., Chundi, P., Gadhamshetty, V. R. & Chilkoor, G. (2021). Deep learning approach to extract geometric features of bacterial cells in biofilms. pp. 359–368. https://doi.org/10.1007/978-3-030-71704-9_23.